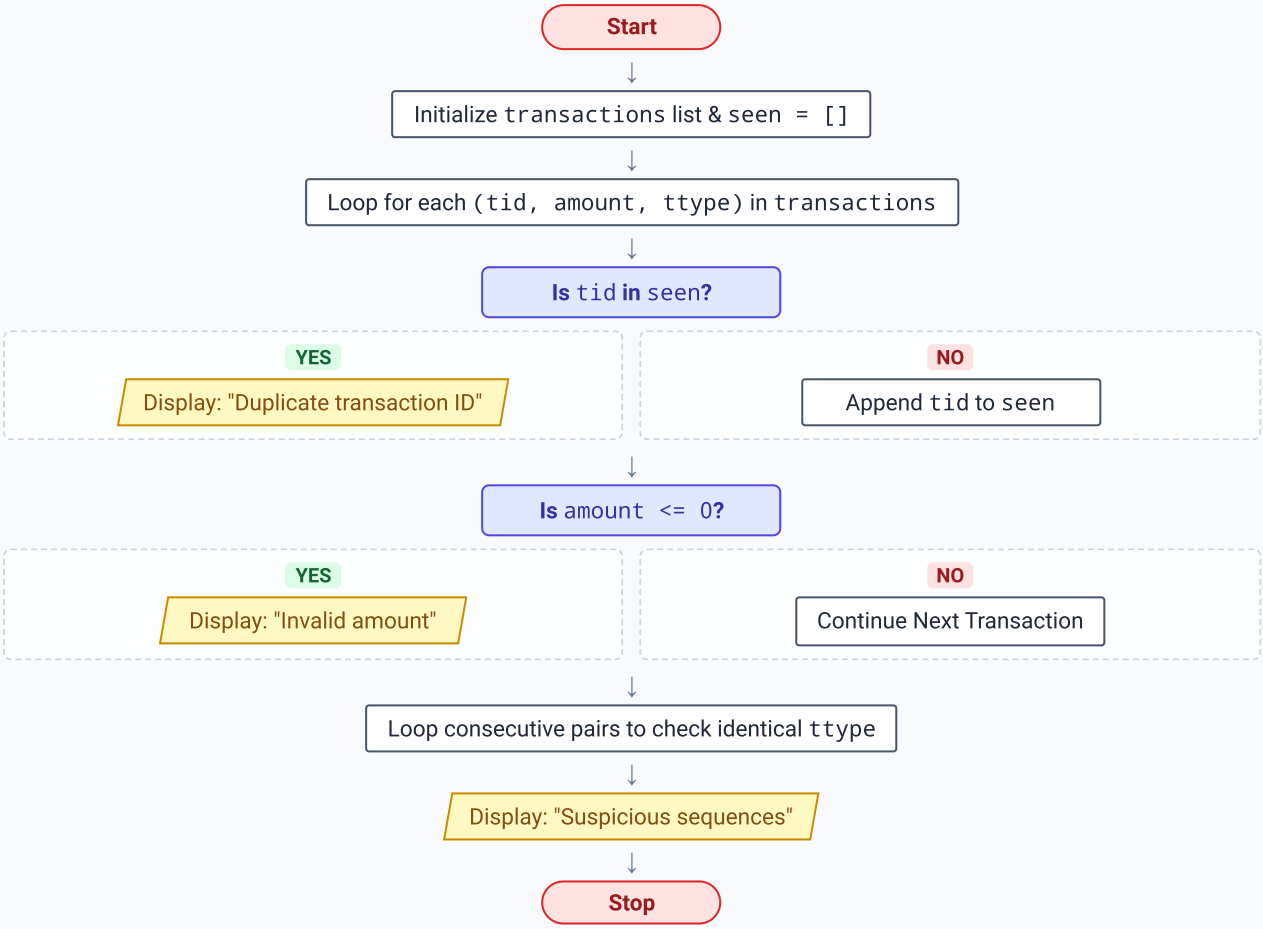
PYTHON FLOWCHARTS GUIDE

Standard Flowchart Logic for Tasks 1 to 10

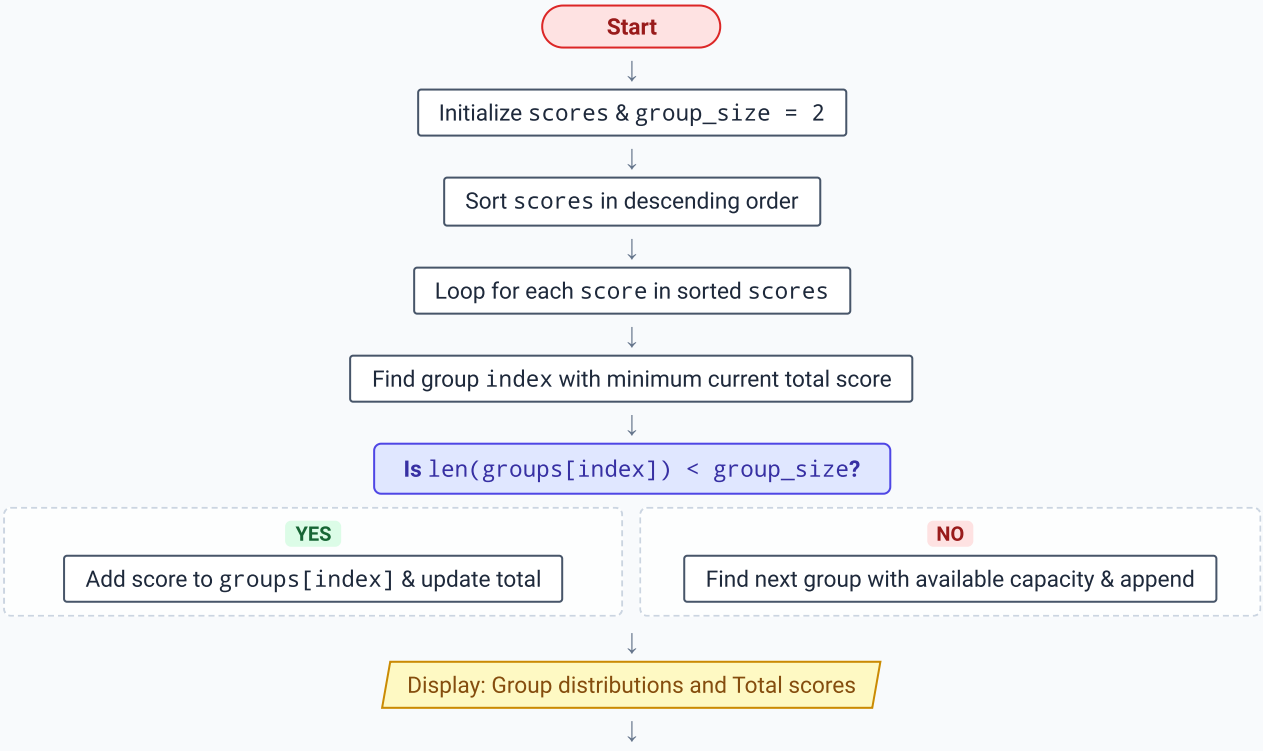
Task 1: Transaction Validation & Duplicate Check

Array / Loop



Task 2: Score Grouping & Balancing

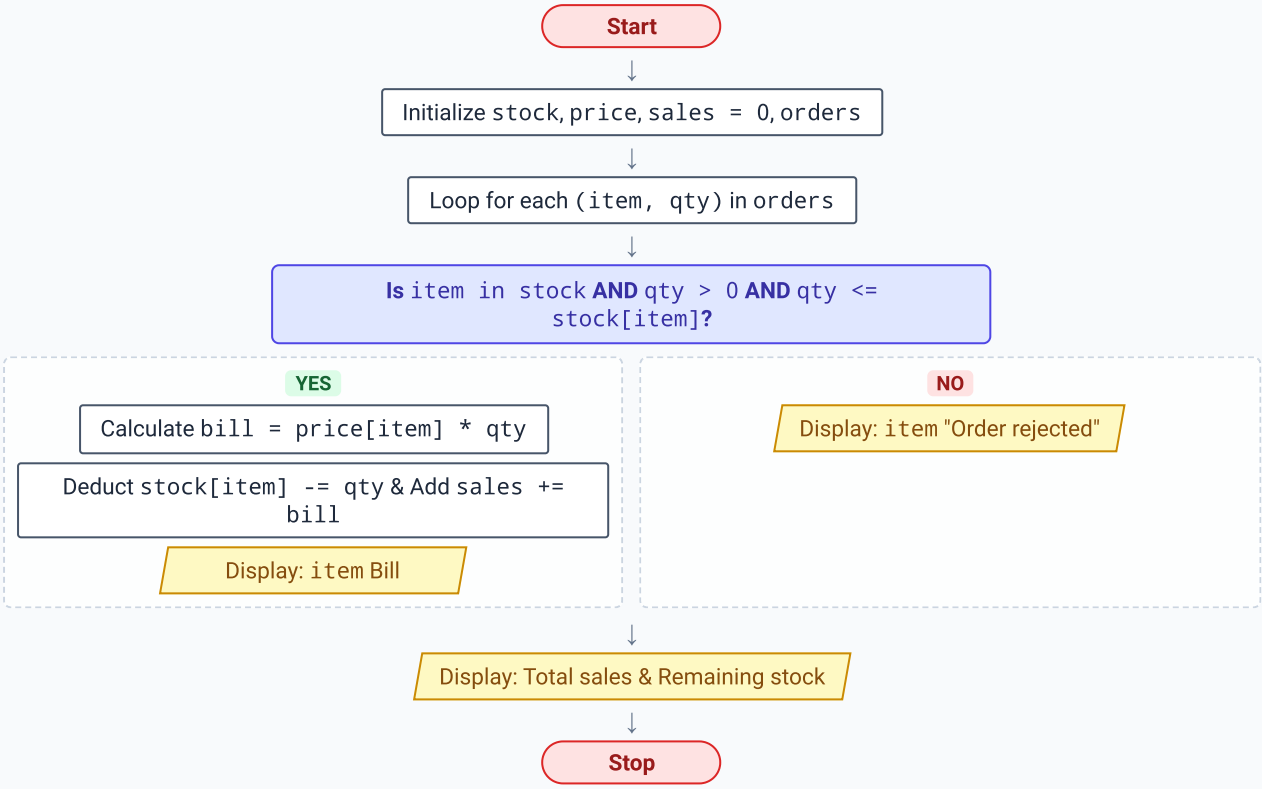
Greedy Distribution



Stop

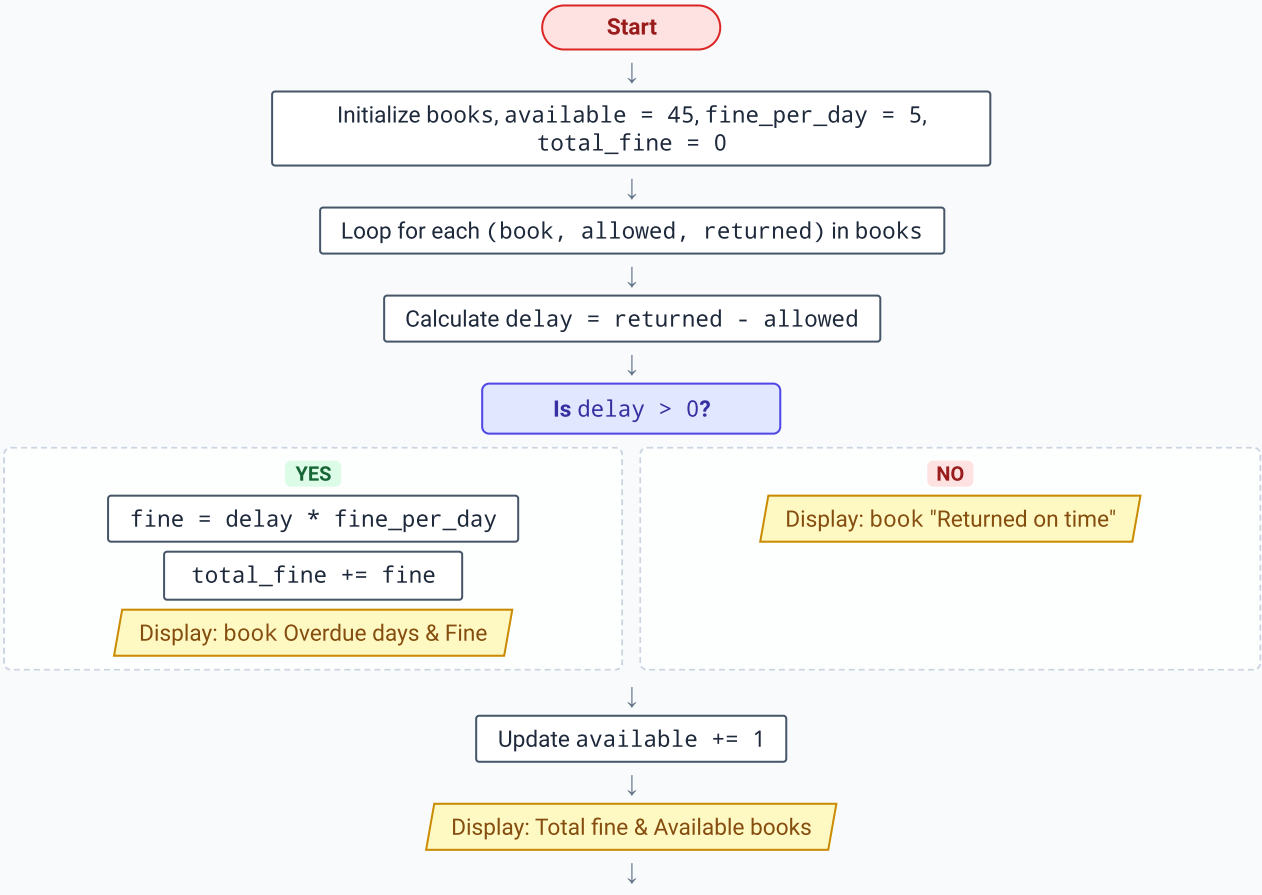
Task 3: Restaurant Order Processing & Inventory

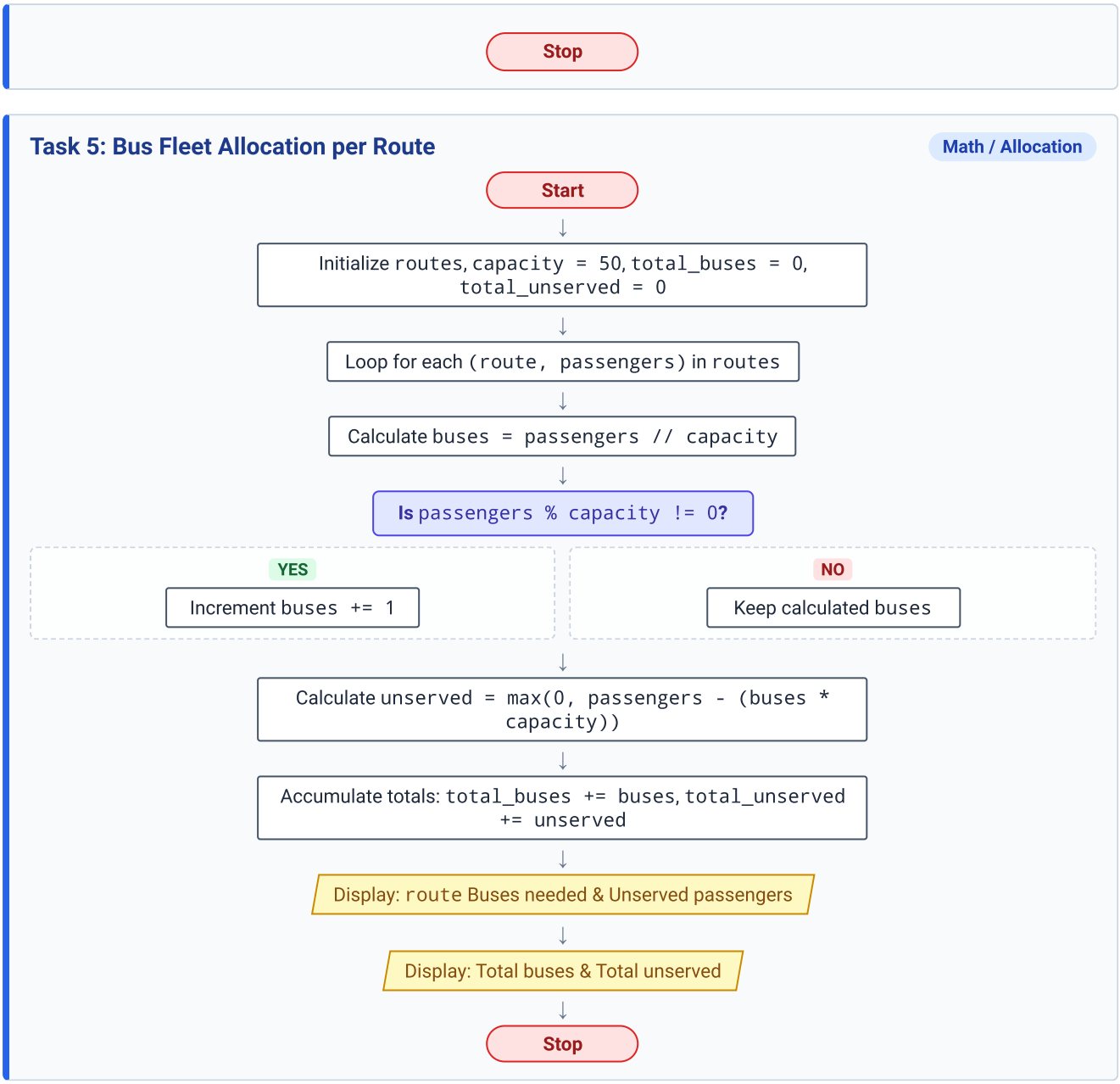
Dictionary / Inventory

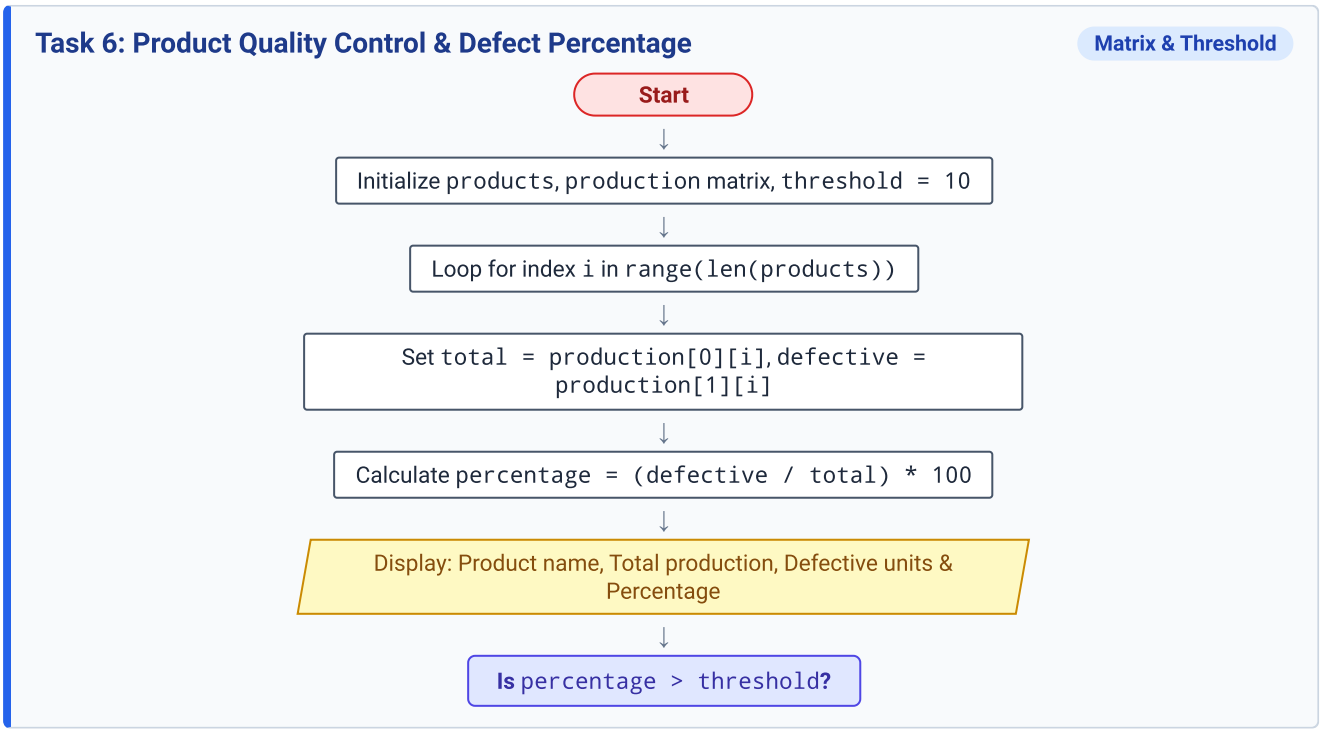


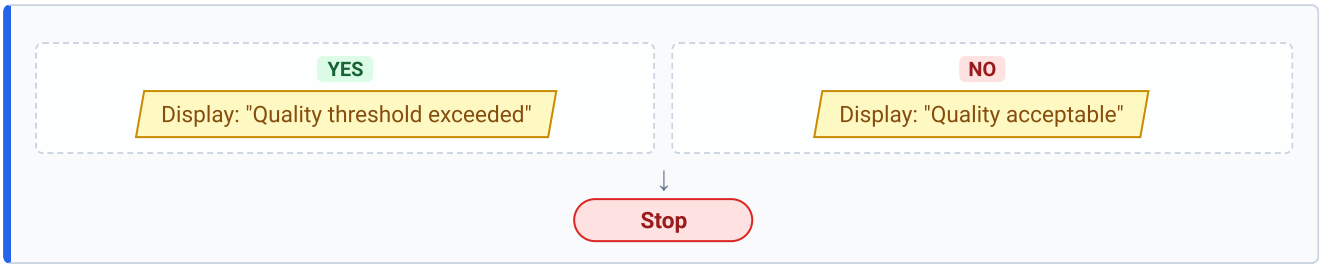
Task 4: Library Book Fine Calculator

Calculation & Counter



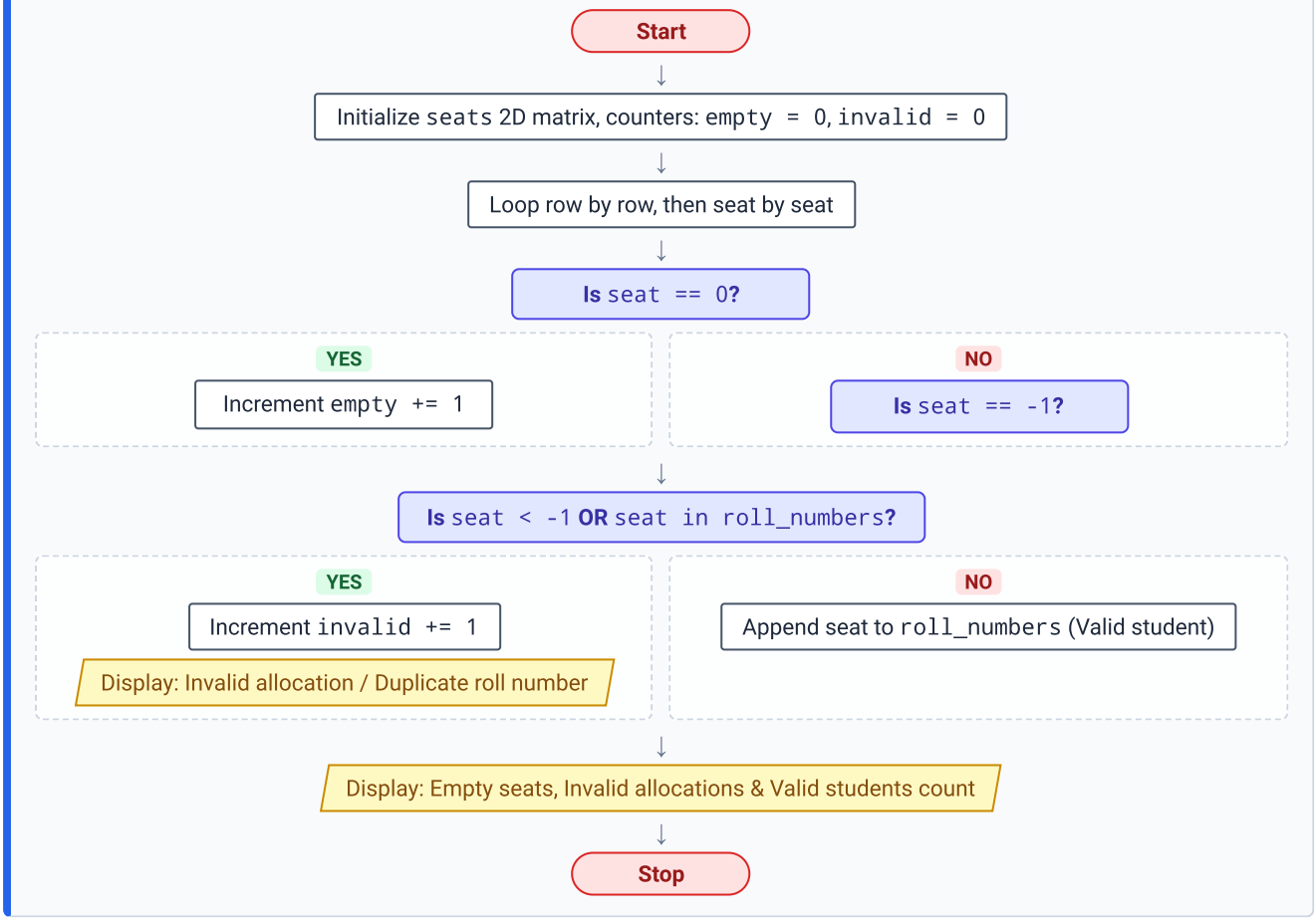

Stop
Task 5: Bus Fleet Allocation per Route
Math / Allocation
Start
Initialize routes, capacity = 50, total_buses = 0, total_unserved = 0
Loop for each (route, passengers) in routes
Calculate buses = passengers // capacity
Is passengers % capacity != 0?
YES
Increment buses += 1
NO
Keep calculated buses
Calculate unserved = max(0, passengers - (buses * capacity))
Accumulate totals: total_buses += buses, total_unserved += unserved
Display: route Buses needed & Unserved passengers
Display: Total buses & Total unserved
Stop


Task 6: Product Quality Control & Defect Percentage
Matrix & Threshold
Start
Initialize products, production matrix, threshold = 10
Loop for index i in range(len(products))
Set total = production[0][i], defective = production[1][i]
Calculate percentage = (defective / total) * 100
Display: Product name, Total production, Defective units & Percentage
Is percentage > threshold?



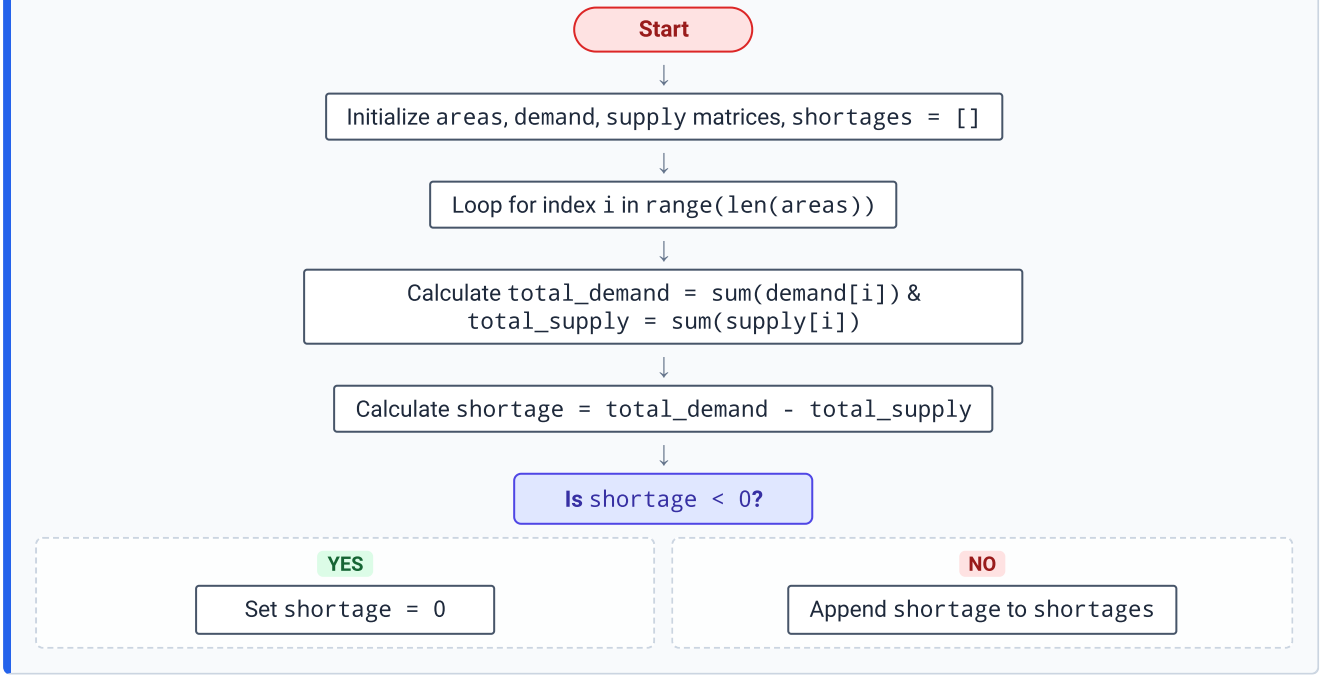
Task 7: Classroom Seat Allocation Validation

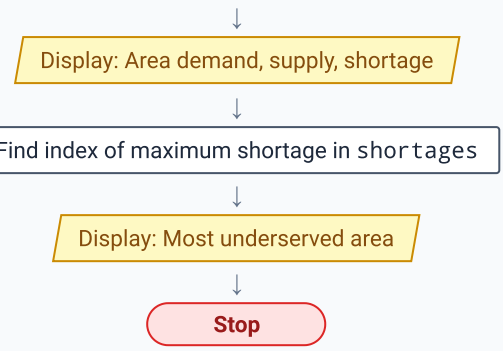
Nested Grid Search



Task 8: Regional Supply & Demand Shortage Analyzer

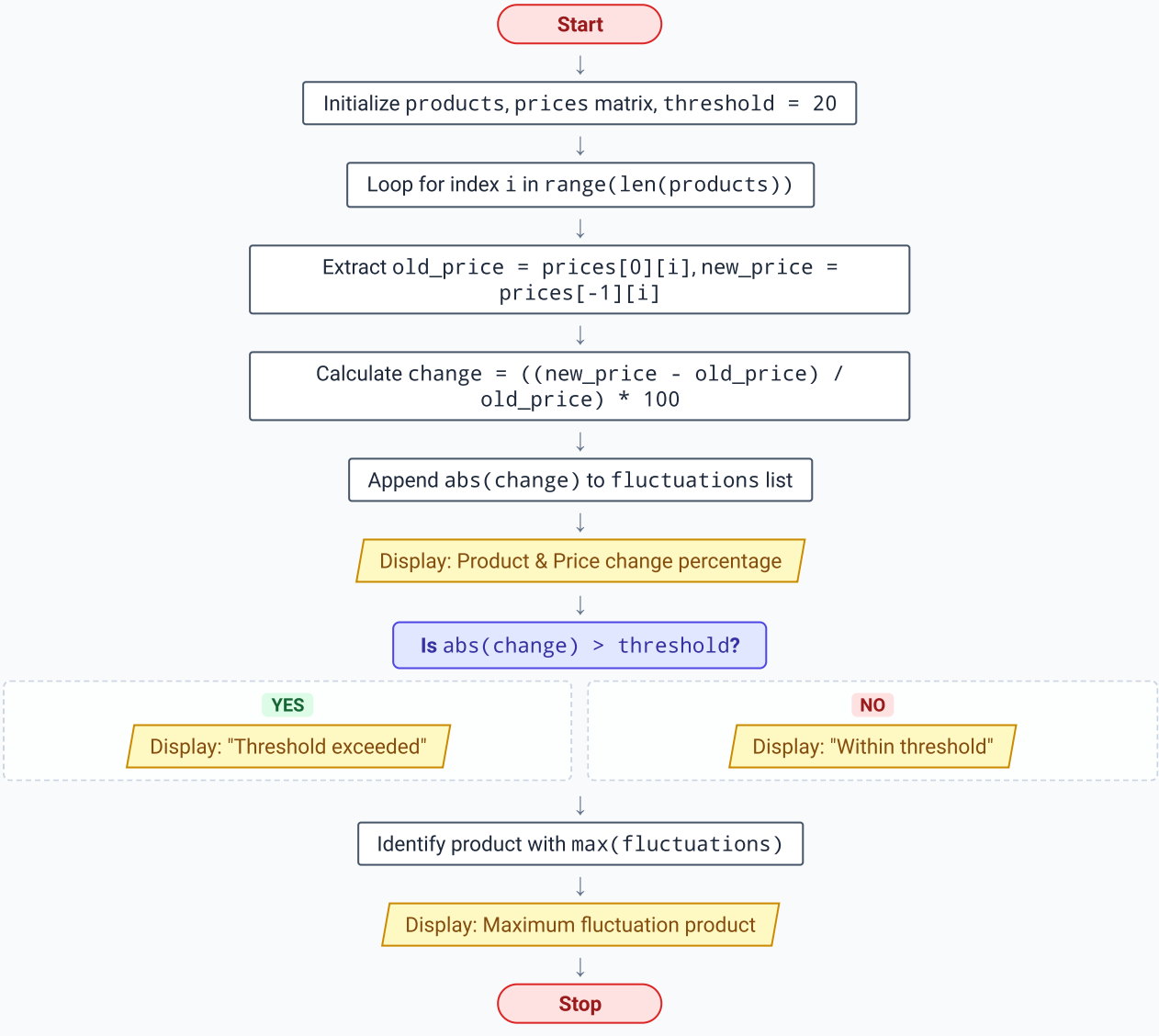
Matrix Summation





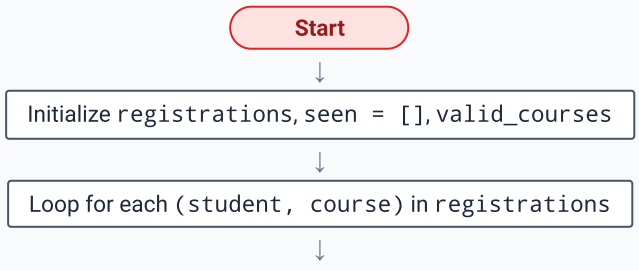
Task 9: Product Price Fluctuation & Threshold Detector

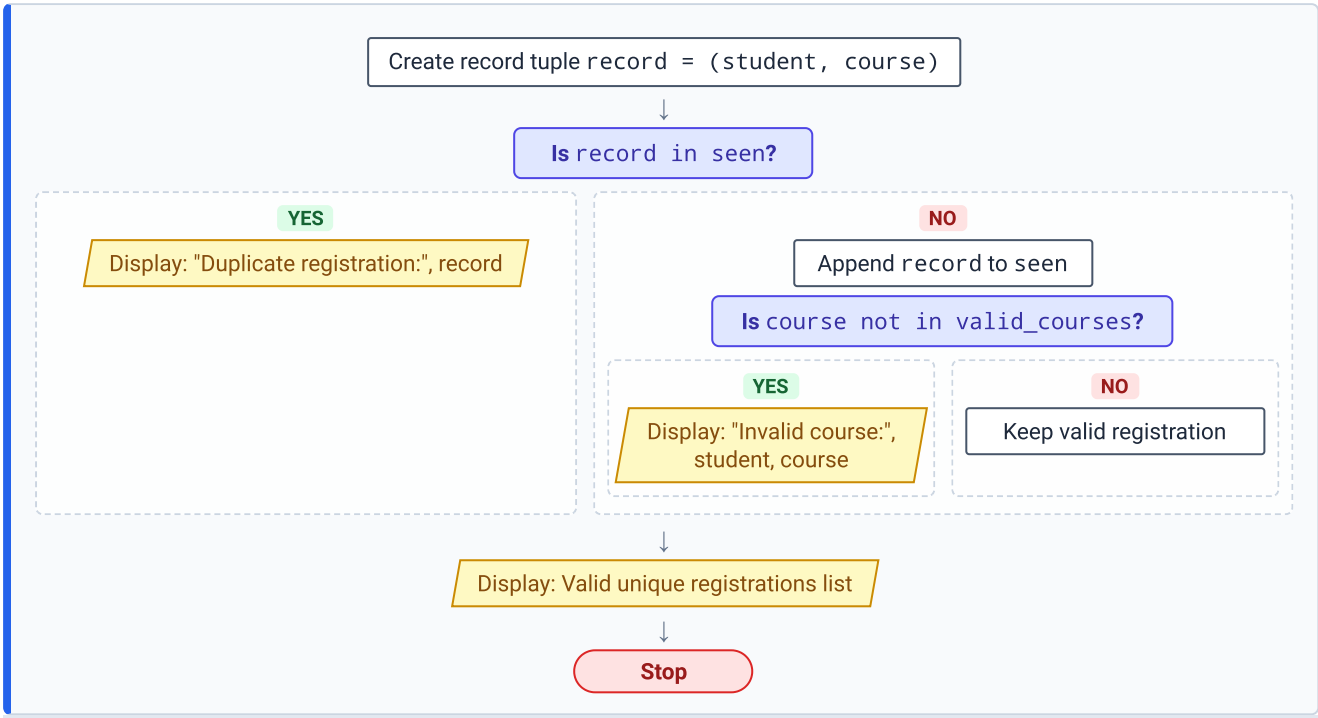
Percentage Change



Task 10: Course Registration Validation & Filter

Tuple & Set Filter




Create record tuple record = (student, course)
Is record in seen?
YES
Display: "Duplicate registration:", record
NO
Append record to seen
Is course not in valid_courses?
YES
Display: "Invalid course:", student, course
NO
Keep valid registration
Display: Valid unique registrations list
Stop